UBL: How Much Is There, and How Fast Is It Moving?

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Introduction: The Question Underneath Everything

There is a very simple question hiding underneath physics: how much is there, and how fast is it moving? We have built an enormous vocabulary around that question, and most of that vocabulary is useful. We talk about mass, energy, fields, particles, waves, time, gravity, temperature, pressure, charge, signal, and noise. But the names can pile up until the physical picture disappears. If we step back from the names for a moment, something much simpler is sitting underneath them. Something is present. It has an arrangement. It moves. When it moves, the arrangement changes.

That is the place I want to start. Not with a list of particles. Not with a list of forces. Not with a stack of equations before we know what the equations are describing. Start with the thing itself. Ask what is there, how it is arranged, how quickly it can change, and what happens to the next piece of the system when it does.

The idea running through this book is that space is not an empty stage with matter placed inside it. The substrate is the physical base layer. Matter is a persistent geometry of that substrate. A wave is organized motion moving through it. What we call a field is not a separate invisible substance laid over the top. It is a way the underlying medium is arranged and moving. Once the same physical medium is allowed to do all of those jobs, many things that look separate begin to look like different conditions of one system.

This does not make mathematics less important. It makes the mathematics easier to see. An equation is useful because it keeps an exact account of physical relationships. The numbers tell us how much, how fast, how far, how dense, how energetic, and how strongly one change is coupled to another. The equations are the bookkeeping. The physical system is what is being counted.

At the smallest scale, this becomes especially interesting. The constants of physics can be combined into a linked set of Planck quantities: a length, a time, a mass, an energy, a density, a force, and a power. Read separately, they look like a collection of extreme numbers. Read together, they look like the dimensions and operating conditions of one base physical scale. The UBL scale takes that idea and treats the Planck condition as the reference against which other physical conditions can be compared.

Then the Balance Laws ask what happens next. If density changes because something moves through a region, and speed changes through that same movement, then density and motion are not separate stories. They are coupled descriptions of one local event. Add geometry, and the

influence of that event changes with distance. Repeat the updates, and patterns can spread, interfere, reinforce, collapse, or settle into stable arrangements.

That is the entire path of this book. The substrate gives us something physical to talk about. The constants give us a scale. The Balance Laws tell us how the local condition updates. Stable geometry gives us matter. Traveling geometry gives us waves. Repeated physical change gives us time. Changed geometry gives us memory. And a sensor, being made from the same physical system, reads both the organized signal and the substrate activity underneath it.

So the question is simple enough to carry all the way through: how much is there, and how fast is it moving? The interesting part is seeing how far that question can take us.

Something Is Always There

Look across an ordinary room. There is a table, a chair, the walls, the air, and what looks like empty space between them. For everyday life, calling that space empty works fine. You can walk through it. You can reach through it. Nothing solid blocks the path. But physical emptiness is a different question from everyday emptiness. Light crosses the room. Radio crosses the room. Gravity is not switched off in the gap between the table and the wall. Put a sensor there and the sensor does not sit at perfect zero waiting for the universe to begin. It already has a baseline.

That baseline is a useful clue. The measuring device is already part of a physical environment before the wanted event arrives. Its atoms are not disconnected from the surrounding space. Its electrons are not in a private universe. The device is already interacting with whatever physical condition fills the region around it. The wanted signal changes that condition, but it does not create the condition from nothing.

I call that underlying condition the substrate. The word matters less than the picture. The substrate is not a fluid poured into a second, emptier space. There is no extra container underneath it. It is the physical base in which distance, motion, fields, and matter exist. When I say substrate, I mean the thing that is already there before we name the particular pattern happening inside it.

This changes the picture of matter. If the substrate is continuous through the region we call empty space and through the regions we call objects, then matter does not need to be a second substance dropped into the first. Matter can be a persistent organization of the same underlying physical medium. The difference between a quiet region and a particle is then not that one contains reality and the other contains nothing. The difference is geometry and activity.

Water gives an easy picture. A vortex behaves like an object. It has a location, a size, an orientation, energy, and a lifetime. It can move through the water and interact with other vortices. But the vortex is not made from special vortex material. It is the water itself organized into a persistent pattern. Remove the pattern and nothing foreign has to be removed from the water. The water simply stops doing that particular thing.

The substrate picture treats matter in the same spirit. A particle is a stable pattern of the underlying medium. A traveling wave is a pattern being passed forward. A stable structure is a pattern whose motion closes into a geometry that can keep rebuilding itself. The medium and the pattern should not be confused, but they also should not be separated into two different realities.

At the smallest scale, the natural question is whether this medium has a minimum working geometry. If change is passed from one region to another, there must be some physical scale at which the ideas of length, action, energy, and propagation meet. That is where the Planck quantities become interesting. They are not interesting merely because the numbers are extreme. They are interesting because the same small set of constants gives a whole family of related physical quantities.

It helps to picture the smallest working regions as cells, not because we need to know their exact shape, but because the word gives us a way to think mechanically. A cell is a local piece of geometry capable of having a condition and passing change to its neighbors. It can be more or less active, more or less compressed, and part of a larger pattern. When many such local regions move together, the pattern can travel. When their relationships close on themselves and remain balanced, the pattern can persist.

This makes a wave straightforward. The wave is not a tiny object that leaves one end of the universe and carries the same chunk of substance all the way to the other end. A change here affects the adjacent condition, that change affects the next one, and the organized geometry advances. The pattern travels because the medium is connected.

Matter is the complementary case. Instead of the organized change simply spreading away, the motion forms a geometry that keeps returning into itself or reinforcing its own structure. The result can have a stable location and repeatable properties. We give it a particle name because the persistent geometry behaves as an object. Underneath the name, the mechanism is still movement and arrangement in the substrate.

This also changes the meaning of interaction. Two pieces of matter are not isolated beads somehow reaching across absolute nothing. They are two persistent patterns embedded in one continuous physical system. Their surrounding geometry overlaps. Their motions enter the same medium. Their effects can propagate because there is something between them capable of carrying change.

Once we stop inserting literal nothing between physical events, the basic questions get simpler. How dense is the local condition? How fast is it changing? What direction does the change move? What geometry can the motion support? Those are questions a mechanical system can answer.

And that takes us straight to the constants. If there is a smallest working geometry, the familiar constants should not merely tell us that our equations become uncomfortable near it. Put them together and they should tell us something about the geometry itself.

Planck Measured the Geometry

Three familiar constants are enough to start the small-scale picture. I am going to keep only a few digits because the point here is the relationship, not a page full of decimals.

$$c \approx 3.00 \times 10^{8}\ \mathrm{m/s}$$

$$h \approx 6.63 \times 10^{-34}\ \mathrm{J \cdot s}$$

$$G \approx 6.67 \times 10^{-11}\ \mathrm{m^3 \cdot kg^{-1} \cdot s^{-2}}$$

The first is a speed. The second is an action. The third is the gravitational or density coupling relationship; in my UST notation I write G = d because I am reading that relationship through density and geometry. Whatever notation we choose, the key point is that these quantities are not random measurements of unrelated things. Combine them and a connected physical scale falls out. Using the conventional reduced Planck constant in the usual formulas gives a Planck length of about this:

$$\ell_p \approx 1.62 \times 10^{-35}\ \mathrm{m}$$

That is an absurdly small distance by human standards, but the scale alone is not the interesting part. The interesting part is what comes with it. The same relationships give a corresponding time:

$$t_p \approx 5.39 \times 10^{-44}\ \mathrm{s}$$

And the connection between the two is almost embarrassingly simple:

$$t_p = \ell_p / c$$

A distance divided by a speed gives an interval. This is where a lot of mystery around time can be removed. Planck time is not a separate substance sitting next to Planck length. It is how long propagation at c takes to cross the Planck length. Something moves through a distance. We count the change and call the count time.

Every clock we have ever built does this. A pendulum swings. A crystal oscillates. An atom cycles. A planet rotates. A pulse travels. The mechanisms are different, but the clock always uses one repeatable physical change to count another physical change. We never put a cup under a clock and collect time. We watch matter move and compare the motion.

That is why I say time is matter moving through itself. The phrase is intentionally plain. The system has geometry. The geometry changes. A repeated change provides a count. A clock is matter using one regular motion as a ruler for other motion. Nothing more exotic has to be inserted between the events.

The same Planck construction keeps going. It gives a Planck mass of roughly 2.18×10^{-8} kilograms and a Planck energy of roughly 1.96×10^{9} joules. It also gives associated density, force, and power scales. We could fill a page with the numbers, but that would hide the thing worth noticing. One

small set of constants is generating a linked length, crossing interval, mass-equivalent, energy, density, force, and power.

That reads differently when the quantities are placed side by side. Instead of a collection of unrelated limits, we are looking at a self-consistent physical scale. There is a characteristic size. There is a characteristic propagation interval. There is a characteristic amount of mass-energy. There is a characteristic density. There is a characteristic force and power. The pieces describe each other.

This is what I mean when I say Planck described the geometry. The constants can be read as measurements of how the base physical condition fits together. c tells us about propagation. h tells us about action through a cycle. G, or d in the UST notation, connects density and geometry. Their combinations give the operating scale that belongs to all three together.

UBL then turns the Planck condition into a ruler rather than leaving it as a remote curiosity. The basic ratio is simple:

$$\mathbf{UBL} = \rho / \rho_p$$

The local density or activity is compared with the Planck reference. The ratio is dimensionless. Instead of carrying an enormous collection of units through every comparison, we can ask where a physical condition sits relative to the common reference.

There are two UBL names that need to stay separate because they refer to two related ideas. Universal Base Layer is the substrate itself, the loose physical base. Universal Baseline Level is the normalized level used to describe where a condition sits on the ruler. In the UBL framing, the quiet universal baseline is around 10^{-123} relative to the Planck reference, while UBL = 1 marks the fully concentrated reference condition.

That enormous range is useful because nature spans an enormous range. The physical system can be very quiet without being nothing. It can be organized into ordinary matter without reaching the fully concentrated condition. It can carry waves, build atoms, make stars, and produce black-hole-scale compression while still being described relative to one ruler.

This also gives us a cleaner way to think about constants. A constant does not have to mean a magical number nailed onto the universe from outside. It can be a stable relationship created by the geometry of the system itself. A circle always carries the same relationship between circumference and diameter because of what a circle is. In the same spirit, a universal physical constant can be read as a relationship that remains fixed because the underlying geometry and propagation rule remain fixed.

Now we have the base material, a ruler, and a propagation scale. The next question is the only one physics ever really gets to ask after that: what happens when the local condition changes?

The Balance Laws

Suppose we stop naming phenomena for a minute and look at one small region of the substrate. There is some local density. Some amount of flow passes through the region. The flow has a speed. If material or activity moves into the region, the local density changes. That is the first update.

$$\mathbf{D_{next} = D + D_flow\ S}$$

D is the density already present. D_flow is the amount moving through the system. S is its speed. The next density is the old density plus the movement carried by flow and speed. The important thing is not the notation. The important thing is the physical statement: density changes because something moves.

Speed updates through that same driver:

$$\mathbf{S_{next} = S + k\ D_flow\ S}$$

The same D_flow S term appears in both updates. That matters. Density is not doing one thing while speed lives in some other mathematical universe. One local event changes both descriptions. The substrate moves, the local amount changes, and the motion responds to the changed condition.

The coupling can be written directly:

$$\mathbf{\Delta S / \Delta D = k}$$

The change in speed tracks the change in density through k. If the reaction is weak, speed changes only a little for the same density change. If it is strong, speed responds more sharply. But the two remain tied because their changes come from the same underlying flow.

This is the part that, to me, makes the system feel physical instead of decorative. We do not need one mechanism for density and another mechanism for speed, followed by a third mechanism to explain why they happen to know about each other. They share the driver because they are two measurements of one event.

Then there is distance. A local disturbance can be strong, but its reach does not stay equally concentrated as it spreads outward. Geometry gives it more area over which to distribute. The distance rule is the familiar inverse-square form:

$$\mathbf{G_{next} = G / r^2}$$

Double the distance and the influence per comparable area falls to one quarter. Triple the distance and it falls to one ninth. We do not need a new personality for the system at each radius. Three-dimensional geometry does the work.

Put the updates together and the basic physical story is almost embarrassingly ordinary. Something moves. Movement changes the local density. The changed density affects continuing movement. The influence of the event weakens with geometric separation.

Now repeat that process. This is where simple local rules can produce complicated-looking behavior. A disturbance can spread away. Two disturbances can overlap and interfere. A change can reinforce the motion that produced it. Another arrangement can work against itself and disappear. Some arrangements can oscillate. Some can collapse into a tighter geometry. Some can return to the same relationship again and again.

Those repeating arrangements are the interesting ones. A physical system cannot maintain every shape we can imagine on paper. The geometry has to close. The motion has to fit the geometry. A returning wave has to meet itself in a way that still works. If it does not, the configuration changes until it finds a permitted form or disperses.

That gives quantization a mechanical home. A standing wave on a string does not choose any arbitrary shape. The length of the string and the propagation condition permit particular modes. The same idea becomes much more powerful when the medium itself has a smallest working geometry. Stable configurations are the ones whose motion and boundary conditions fit.

From that viewpoint, a quantum state does not have to begin as an abstract command saying nature is quantized because that is the rule. The allowed state is the geometry the system can maintain. Different stable arrangements have different energy, action, orientation, and response because they are different geometries of the same moving substrate.

A particle is then a very small persistent mode. An atomic arrangement is a larger stable relationship among persistent modes. A molecule is a shared geometry that can hold several atomic structures together. At every step the names become more specialized, but the foundation stays the same: local amount, local motion, coupling, distance, and geometry.

This is why the Balance Laws make the quantum world look less mysterious. Quantum physics can be mathematically difficult because complicated geometries have complicated behavior. That is not the same as saying the underlying mechanism has to be complicated. A coastline can have an impossibly complicated boundary while still being made by ordinary water, rock, wind, and time. Complexity in the result does not require a different fundamental mechanism for every bend in the line.

The Balance Laws are deliberately small. That is their point. They reduce the local update to density and speed riding on the same flow and the global reach to geometry with distance. If a large family of physical behavior can be built from those relationships, then the unity is not in the vocabulary. It is in the repeated mechanism.

And now we can take that mechanism and look again at the names we use every day: wave, particle, heat, time, field, and memory.

The Different Names We Give the Same Motion

Start with a wave. A wave is organized change traveling through a medium. That statement is so ordinary with water and sound that we barely think about it. The shape moves because neighboring parts of the medium keep changing one another. The wave is real, but it is not a second material floating above the medium. It is what the medium is doing.

Carry that picture down to the substrate and a field disturbance becomes easier to visualize. The local state changes, neighboring regions respond, and the pattern propagates. We can describe the pattern with frequency, wavelength, phase, amplitude, energy, or momentum. Those are measurements of the organized motion. They do not require the wave to become a separate substance.

Matter is what happens when organized motion does not simply pass through and leave. The geometry closes into a stable pattern. The structure persists because its internal movement continually rebuilds the configuration. That is why the same object can show wave behavior and particle behavior. It is a localized geometry built from wave-like motion. The old argument about whether it is really a wave or really a particle starts from a false choice. It is a stable moving geometry, so of course both descriptions can appear.

Heat fits the same picture. Temperature is not a separate fluid hidden inside matter. It reflects physical motion and the distribution of energy among the available structures and modes. At ordinary scales we often describe atoms and molecules jiggling. In a substrate picture, that visible motion sits on top of and interacts with the deeper activity of the medium. The amount and pattern of movement determine the thermal condition we measure.

Pressure is another way motion and geometry show themselves. Movement striking or loading a boundary produces a measurable push. Compress the available geometry and the relationships among the moving parts change. Again, nothing fundamentally new had to be inserted. We changed the amount, movement, and arrangement and gave the resulting measurement a useful name.

Electromagnetism can be read the same way. Instead of imagining an abstract field as a second reality surrounding matter, treat EMF as organized activity of the substrate. Charge and magnetic behavior then describe particular asymmetries and motions of the geometry. The mathematical field remains essential because it tells us how that condition varies through space, but the physical picture underneath it is motion and arrangement rather than an invisible bookkeeping substance.

Gravity belongs in the same family at a different scale. Concentrated matter changes the surrounding condition, and that change has geometry and distance. The inverse-square relationship already tells us that reach is shaped by space. In the UBL picture, gravity is the large-scale behavior of density and substrate geometry rather than a separate mechanism that suddenly appears when objects get big enough.

Time returns here because all of these named phenomena are changes. We measure a wave because a detector changes. We measure heat because a physical system changes. We measure

motion by comparing positions at different counted physical states. A clock is another changing system used as the comparison. Time is the common bookkeeping language we attach to the sequence.

Memory sounds like it belongs in a completely different book until we ask what memory has to be physically. Something happened before, and because it happened before, the system responds differently now. For that to be true, the earlier event must have left a physical difference behind.

That difference is geometry. A signal moves through a system. The EMF and local physical motion alter the arrangement. The immediate signal can pass, but the changed arrangement remains. The next signal does not arrive at the old system. It arrives at a system whose geometry has been modified by the previous event.

The EMF shapes the geometry. The geometry is the memory.

This principle does not require a brain. Bend a piece of metal and the changed shape records the stress history. Magnetize a material and the alignment carries information about the earlier field. Change a biological connection and later activity follows a different physical path. The complexity of the storage can become enormous, but the mechanical idea is simple: previous movement changed the present geometry.

That gives us a useful way to think across scales. A stable particle carries geometry. An atom carries geometry. A molecule carries geometry. A solid carries geometry. A living system constantly changes geometry while maintaining enough persistence to remain the same organized system. The history of the motion becomes part of the structure that future motion encounters.

Once this is clear, many nouns begin turning back into verbs. A field is the substrate in a particular condition. A wave is the substrate propagating change. Matter is the substrate maintaining geometry. Time is change being counted. Memory is earlier change preserved in present geometry.

The point is not to get rid of the old names. We need them. The point is to remember what they are names for. If the underlying mechanism is amount, motion, coupling, and geometry, then different branches of physics can use different vocabulary without requiring different universes underneath them.

That brings us to measurement, because measurement is where all of these ideas have to meet a real piece of hardware. A sensor is matter. It is geometry in the substrate. It cannot stand outside the system and look in. Whatever the substrate is doing, the sensor is already participating in it.

Signal, Noise, and the UBL Scale

Put a sensor on a bench and do nothing to it. Do not transmit the wanted signal. Do not deliberately excite the system. The output still moves. There is a baseline. Electronics gives us several useful names for contributions to that baseline: thermal noise, shot noise, interference, device noise, quantization effects, and more. Those labels help engineering. But underneath the labels is a physical fact that should not be overlooked: the sensor is already interacting with its environment before the wanted signal arrives.

That sounds obvious when stated plainly, but it changes how we think about measurement. The baseline is not automatically meaningless garbage surrounding the real physics. The baseline is itself a physical measurement. Some of it may be unwanted for the particular question we are asking, but unwanted and unreal are not the same thing.

The sensor is made of the same physical system it is trying to observe. Its conductors, junctions, atoms, charge distributions, and ADC are not outside the substrate. The surrounding activity continuously couples into the hardware. When an organized signal arrives, it rides into a system that was already moving.

So the clean distinction is not between real signal and fake noise. It is between motion we have resolved into the pattern we are looking for and motion that remains unresolved with respect to that pattern.

In the UBL notes I write the relationship this way:

$$\alpha = \mathbf{h} - \boldsymbol{\mu}$$

Here h is total mass-motion, μ is unresolved mass-motion, and α is the resolved action left after the unresolved part is separated. In plain language, everything is moving. Some of that movement belongs to the organized event we can identify. Some belongs to the background and other motion we have not classified as part of the event.

This is why I do not like saying that noise is simply removed. You can remove numbers from a data stream. You can filter frequencies. You can average samples. But physically, the activity did not stop existing because an algorithm decided it was inconvenient. A better measurement starts by accounting for the baseline rather than pretending the baseline is nothing.

The UBL scale supplies the reference. The Planck condition is the upper reference. The universal baseline sits at the opposite end of the enormous density ratio, around 10^{-123} in this framework. A local measurement sits somewhere on that same scale. The purpose of the ratio is not to make every sensor display a cosmic number. The purpose is to give the physical condition and the measured deviation a common ruler.

For digital measurement, the ADC is the natural place to care about that relationship because that is where the physical voltage, current, field response, or other analog quantity becomes a numerical representation. Once the conversion has happened, every later operation is working on

those numbers. If the physical baseline is misread at the conversion point, later computation can become an elaborate attempt to repair a bad reference.

In the UBL correction approach, the 10^{123} relationship is applied at the ADC so the baseline scale is accounted for at the point where the signal becomes a count. The intention is not to erase the substrate. It is to keep the unresolved baseline and the resolved deviation in the correct relationship before downstream processing begins.

This is where the physics and the engineering reconnect. The same substrate picture that says there is no empty space also says there is no perfectly isolated sensor. The same Balance Laws that describe local density and speed changes say the detector's local condition must respond to incoming flow. The same UBL ruler used to compare physical density conditions can be used as the reference when the measurement is digitized.

Seen this way, signal and noise are not enemies. They are classifications. The signal is organized motion that matches the event we are trying to read. The noise band is unresolved motion in the same physical system. Improving the measurement means distinguishing those relationships more accurately, not declaring one part of reality unphysical.

That also explains why brute force has limits. More samples, more filtering, more compute, and more elaborate models can be extremely useful, but computation does not get to change what entered the ADC. If the original numerical representation confuses the baseline relationship, downstream sophistication is still operating on that representation. Fixing the reference at the conversion point attacks the problem earlier.

Now step back from the sensor and look at the whole picture we have built. The substrate supplies the physical medium. The constants supply the base scale. Planck geometry gives us length, motion interval, energy, density, force, and power as connected quantities. UBL provides a common ruler. The Balance Laws update density and speed through the same local driver and let geometry control reach with distance.

A traveling disturbance is a wave. A persistent disturbance is matter. Stable geometries at the smallest scale appear as quantized states because not every configuration can maintain itself. Electromagnetic behavior is organized local substrate motion. Gravity is large-scale density and geometric behavior. Time is the count of physical change. Memory is previous motion preserved in geometry. Noise is unresolved movement. Signal is resolved movement.

The vocabulary is still useful because the universe builds structures at wildly different scales. Nobody wants to describe a radio circuit, a molecule, a memory, and a star using the same paragraph every time. Special words let us work efficiently. But underneath the special words we do not need a new physical universe for every subject.

We keep coming back to the same questions. How much is there? How is it arranged? How fast is it moving? How does that movement change the arrangement? How far does the influence reach? What geometry survives the next update?

That is the reason the UBL scale and the Balance Laws fit together. One tells us where the condition sits. The other tells us how the condition moves. Together they turn a pile of separate nouns back into one moving physical system.

Conclusion: That Is Physics

We can finish where we started because the question did not get more complicated as we went. The examples got more complicated. The geometry got more complicated. The names got more complicated. The basic physical question stayed the same.

How much is there? How fast is it moving? Where is it concentrated? What shape does the motion take? What does that shape do to the next movement?

The substrate gives us something physical that can have those properties. The constants describe fixed relationships within that physical system. Put the constants together and the Planck scale appears, not as a lonely collection of extreme numbers but as a connected set of length, propagation interval, mass-energy, density, force, and power relationships.

The UBL scale turns that base condition into a ruler. We can describe a quiet condition and a concentrated condition on the same scale. We do not have to invent a new reference every time the density changes by another fifty orders of magnitude.

The Balance Laws tell us what movement does locally. Flow changes density. The same flow changes speed. The two changes remain coupled. Distance changes the reach through geometry. Repeat those updates and the system can spread, interfere, reinforce, oscillate, collapse, or settle into stable structures.

Those structures are the things we normally introduce first and explain later. Here they arrive later because they are results. Matter is persistent geometry. A wave is traveling geometry. A quantum state is an allowed stable geometry. Time is movement counted against movement. Memory is earlier movement preserved as changed geometry. Signal is organized movement we have resolved. Noise is movement we have not resolved into the event we are trying to measure.

That is what unification looks like to me. It does not mean using one word for everything. It means the mechanism does not have to be replaced every time the scale changes. The same physical system keeps moving. The same geometry keeps constraining the movement. The same relationships keep giving us the measurements. Physics is the bookkeeping of that moving universe. The equations keep the accounts straight, but the physical story underneath them is wonderfully plain. How much is there, and how fast is it moving? That is physics.

Equation Reference

Speed of light: $c \approx 3.00 \times 10^{8}$ m/s

Planck constant: $h \approx 6.63 \times 10^{-34}$ J·s

Gravitational / density coupling: $G \approx 6.67 \times 10^{-11}$ $m^3 \cdot kg^{-1} \cdot s^{-2}$

Planck length: $\ell_p \approx 1.62 \times 10^{-35}$ m

Planck time: $t_p \approx 5.39 \times 10^{-44}$ s

Planck crossing relationship: $t_p = \ell_p / c$

Universal Baseline Level: $UBL = \rho / \rho_p$

Density update: $D_{next} = D + D_flow\ S$

Speed update: $S_{next} = S + k\ D_flow\ S$

Density-speed coupling: $\Delta S / \Delta D = k$

Distance law: $G_{next} = G / r^2$

Resolved action: $\alpha = h - \mu$

Universal Base Layer: the physical substrate. Universal Baseline Level: the normalized position on the UBL scale. In this framework the quiet universal baseline is approximately 10^{-123} relative to the Planck reference, and UBL = 1 is the fully concentrated reference condition.

How much stuff is there? + How fast is it moving?